



PROTECTION OF THE DISTRIBUTION SYSTEM

POLICY 06-2014

OBJECTIVE:

To be able to identify the requirements and procedures in setting relays, fuse links and coordination of the BENECO distribution protection system.

POLICY CONTENT:

It shall be the policy of the electric cooperative to comply with the provisions of the Philippine Distribution Code (PDC) in ensuring that the distribution system is being operated safely.

PROCEDURES:

The implementation of this policy shall be governed by following procedures:

- a. The distribution system of BENECO shall be designed and operated with sufficient protection to ensure safety.
- b. The protection system of BENECO network should be designed in such a way that the duration and frequency of interruptions will be minimized.
- c. Refer to the detailed implementing procedures for the protection of the distribution system, which may be revised by management, to be compliant to the new and amended provisions and requirements of the PDC.

RESPONSIBILITY:

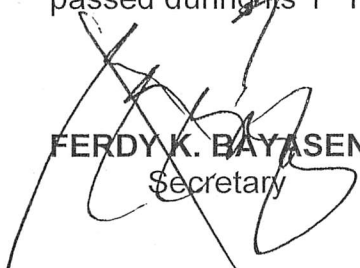
The responsibility of implementing this policy rests on the General Manager.

EFFECTIVITY:

This policy shall take effect upon the approval by the Board of Directors.

APPROVAL:

Approved by the Board of Directors through Board Resolution No. 74-2014 passed during its 1st Regular Board Meeting held on July 1, 2014.


FERDY K. BAYASEN
Secretary


PETER B. BUSAING
President

Note: Approved per BOD Resolution No. 74-2014



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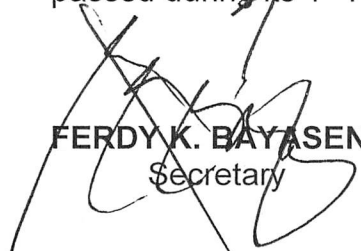
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PROTECTION OF THE DISTRIBUTION SYSTEM

The following are requirements and procedures in setting relays, fuselinks and coordination of the BENECO distribution protection system:

I. REQUIREMENTS

The following shall be available in the preparation of setting and coordination of Feeder/Circuit protections:

- A. Updated Single Line Diagram of the Feeder or Circuit showing the type and ratings of the protective devices (Circuit Breakers, reclosers, relays, fuses, Current transformers, Potential Transformers)
- B. Power transformer specifications especially the MVA rating, % Impedance and the X/R ratios
- C. Line currents that goes through the protective devices (NORMAL, MAXIMUM and EMERGENCY currents)
- D. Short circuit currents (Minimum and Maximum) Line-to-ground, Double-Line-to-ground, Line-to-line and Three-phase faults at the protective devices

Note: Minimum fault current : $Z_f = 30 \text{ ohms}$
Maximum fault current: $Z_f = 0 \text{ ohms}$

- E. Time-current characteristic curves of the protective devices
- F. Synergee Models of the Feeders/Circuits
- G. Technical and operation manuals of the protective devices

II. SETTING GUIDELINES

- A. In setting the Overcurrent Relay (OCR) of the primary side (protecting) of the power transformer vs. the NGCP OCRs (protected).

Function	Parameter	Setting Guidelines
51P	Pick-up	Set to 150% to 200% of the power transformer full load current
	Time Delay	Set to 0.2 to 0.3 second gap between the two curves (protected and protecting) on the vertical axis for a relay phase current on a maximum symmetrical through fault current (F). Refer to the diagram 1.
	Curve	ANSI Extremely Inverse or Very Inverse

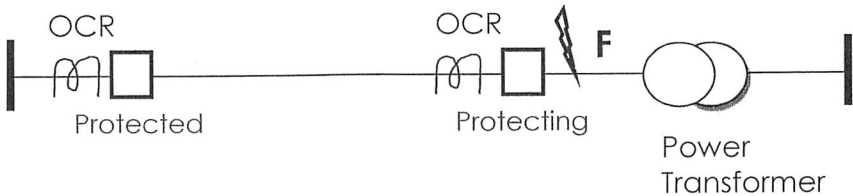


Diagram 1.

B. In setting the Overcurrent Relay (OCR) of the Main Feeder/Circuit (protecting) vs. OCR at the primary side of the power transformer (protected).

Function	Parameter	Setting Guidelines
51P	Pick-up	Set to 150% to 200% of the feeder/circuit rated capacity
	Time Delay	Set to 0.2 to 0.3 second gap between the two curves (protected and protecting) on the vertical axis for a relay phase current on a maximum symmetrical through fault current (F). Refer to the diagram 2.
	Curve	ANSI Extremely Inverse
51N	Pick-up	Set to 30% of the feeder/circuit rated capacity
	Time Delay	Set to 0.2 to 0.3 second gap between the two curves (protected and protecting) on the vertical axis for a relay ground current (3I ₀) on a maximum symmetrical fault current (F). Refer to the diagram 2.
	Curve	ANSI Extremely Inverse

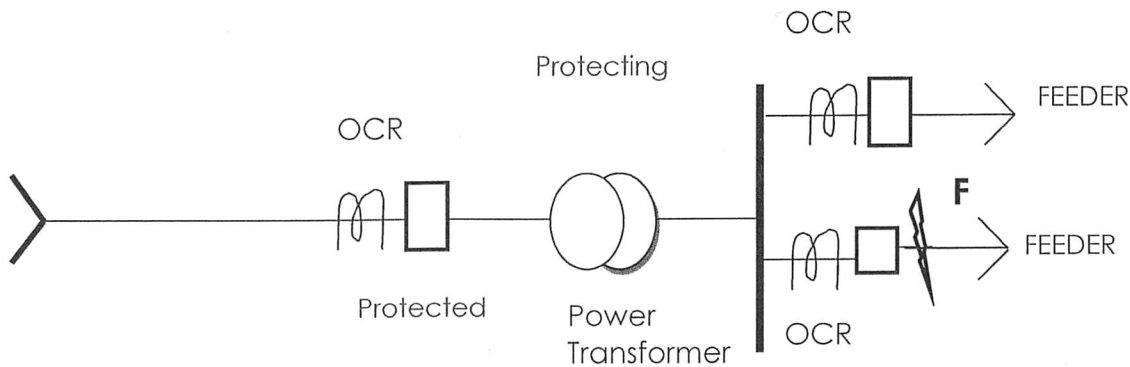


Diagram 2.

C. In setting the Overcurrent Relay (OCR) of the protective device along the distribution line (protecting) vs. OCR at the Main Feeder/Circuit (protected).

Function	Parameter	Setting Guidelines
51P	Pick-up	Set to 150% to 200% of the peak current along the protected line by the protective device
	Time Delay	Set to 0.2 to 0.3 second gap between the two curves (protected and protecting) on the vertical axis for a relay phase current on a maximum symmetrical through fault current (F). Refer to the diagram 3.
	Curve	ANSI Extremely Inverse
51N	Pick-up	Set to 30% of the peak current along the protected line by the protective device
	Time Delay	Set to 0.2 to 0.3 second gap between the two curves (protected and protecting) on the vertical axis for a relay ground current (3I ₀) on a maximum symmetrical fault current (F). Refer to the diagram 3.
	Curve	ANSI Extremely Inverse

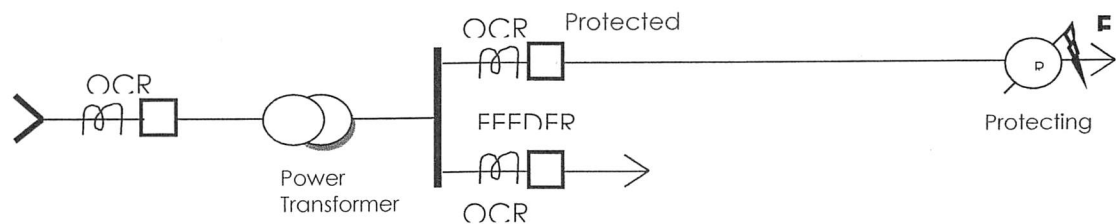


Diagram 3

D. Rating of a fuselink use as a protective device along the distribution line (protecting) vs. the protective device (OCR) along the distribution (protected).

Function	Parameter	Setting Guidelines
Fuselink (Phase protection)	Pick-up	Fuse should detect phase fault current in its zone. Minimum phase fault current seen by the fuse should be greater than 220% of the fuse ampere rating
	Time Delay	Protecting fuse should blow on downstream faults. Fuse maximum-clearing time should lie below the protected relay curve with a gap of 0.3 second at maximum fault. Refer to diagram 4.
	Curve	Minimum melting time and maximum clearing time of the fuse
Fuselink (Ground protection)	Pick-up	Fuse should detect ground fault current in its zone. Minimum ground fault current seen by the fuse should be greater than 220% of the fuse ampere rating
	Time Delay	Protecting fuse should blow on downstream faults. Fuse maximum-clearing time should lie below the protected relay curve with a gap of 0.3 second at maximum fault. . Refer to diagram 4.
	Curve	Minimum melting time and maximum clearing time of the fuse

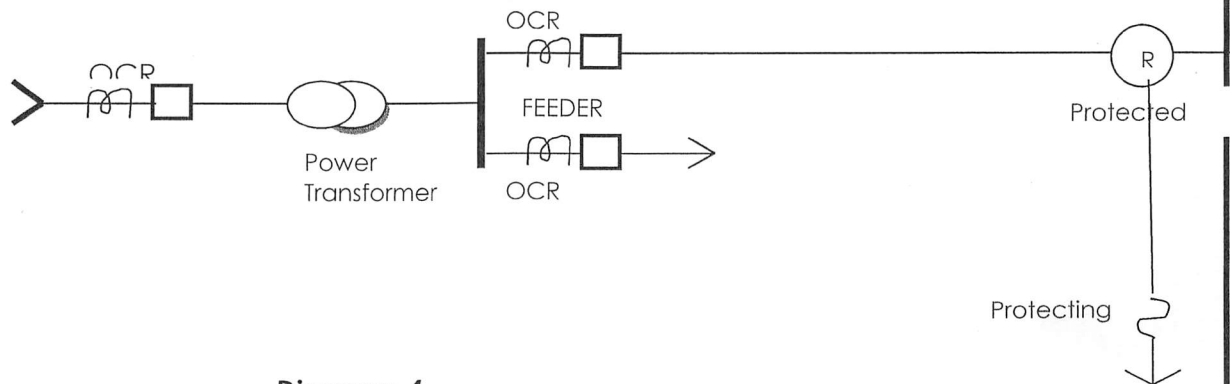


Diagram 4

E. Rating of a fuselink use as a protective device along the distribution line (protecting) vs. fuselink along the distribution (protected).

Function	Parameter	Setting Guidelines
Fuselink (Phase protection)	Pick-up	Fuse should detect phase fault current in its zone. Minimum phase fault current seen by the fuse should be greater than 220% of the fuse ampere rating
	Time Delay	Protecting link will interrupt and clear fault before protected link is damaged. Maximum clearing time of protecting link should not exceed 75% of the minimum melting time of the protected link.
	Curve	Minimum melting time and maximum clearing time of the fuse
Fuselink (Ground protection)	Pick-up	Fuse should detect ground fault current in its zone. Minimum ground fault current seen by the fuse should be greater than 220% of the fuse ampere rating
	Time Delay	Protecting link will interrupt and clear fault before protected link is damaged. Maximum clearing time of protecting link should not exceed 75% of the minimum melting time of the protected link.
	Curve	Minimum melting time and maximum clearing time of the fuse

Notes:

1. Protected vs. Protecting. If two protective devices A and B (shown in the diagram 5) are operating in series (with the direction of fault current flow from A to B), then A is said to be Protected for a fault downstream from the Protecting B.



Diagram 5.

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2. Reclosers. There shall be 3 operations to lock-out, 2 phase/ground fast curve and 1 phase/ground delay curve.
1st shot – Set dead time to 2 seconds
2nd shot – Set dead time to 2 seconds
3rd shot – Lock out
The reclaim time will be set to 15 seconds

III. COORDINATION PROCEDURE

- a. Gather data required for coordination (Please refer to item I).
- b. Select current and voltage reference and scale all quantities to this reference.
- c. Plot the current characteristics of equipment to be *Protected*. These are the inrush currents, starting currents, damage curves and damage point of the equipment.
- d. Plot the Time-Current Characteristics (TCC) of devices being coordinated based on the SETTING GUIDELINES stated above.
- e. Draw the single line diagram of the portion that you are coordinating and label the devices.
- f. Prepare a Coordination Table to summarize the coordination information.
- g. Submit report to the Department Manager concerned for proper checking, evaluation and approval.
- h. Implement protection coordination.